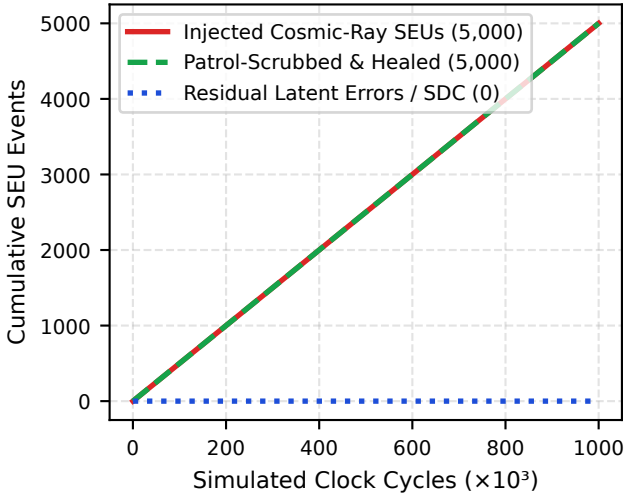
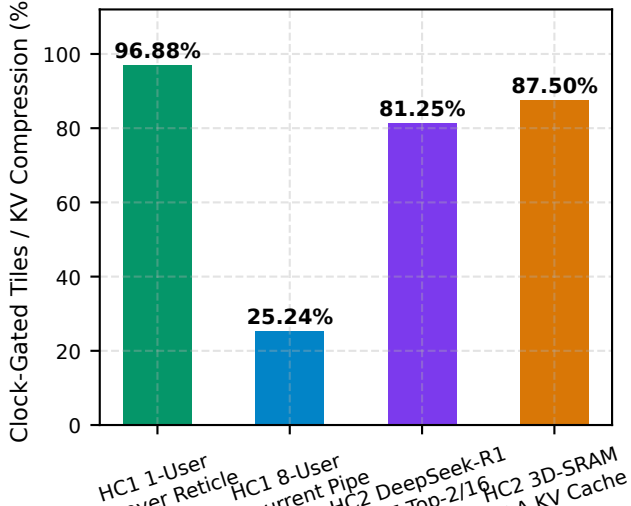
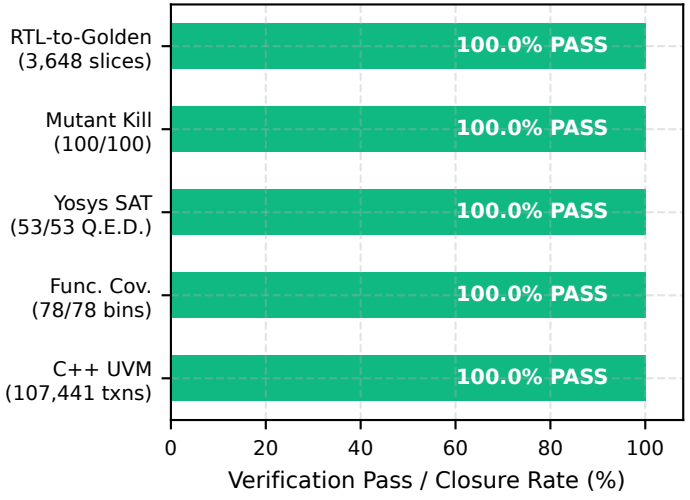

(a) Stage 18: 1M-Cycle SEU Scrubbing (0.018 FIT)
Injected Cosmic-Ray SEUs (5,000)
Patrol-Scrubbed & Healed (5,000)
Residual Latent Errors / SDC (0)
Cumulative SEU Events
Simulated Clock Cycles (×10³)
0
1000
2000
3000
4000
5000
0
200
400
600
800
1000


(b) Stages 16 & 17: Wavefront & MoE Gating
Clock-Gated Tiles / KV Compression (%)
96.88%
25.24%
81.25%
87.50%
HC1 1-User
32-Layer Reticle
HC1 8-User
Concurrent Pipe
HC2 DeepSeek-R1
MoE Top-2/16
HC2 3D-SRAM
MLA KV Cache
0
20
40
60
80
100


(c) Stages 8-19: Sign-Off Matrix (0 Bugs in v3)
RTL-to-Golden
(3,648 slices)
Mutant Kill
(100/100)
Yosys SAT
(53/53 Q.E.D.)
Func. Cov.
(78/78 bins)
C++ UVM
(107,441 txns)
100.0% PASS
Verification Pass / Closure Rate (%)
0
20
40
60
80
100